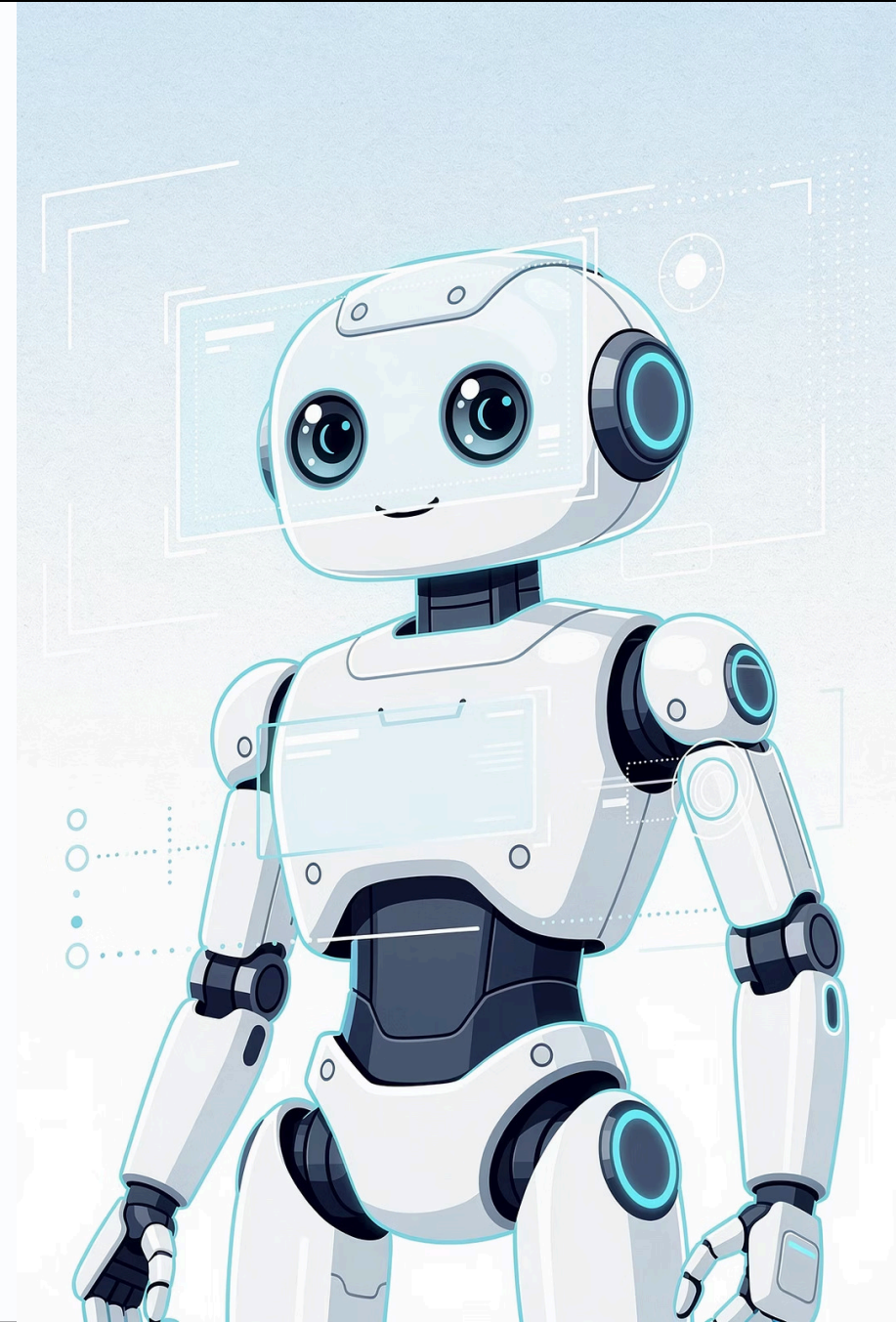


Session 15:

AI That Sees - Computer Vision

Today, you'll discover how AI "sees" the world. We will look under the hood at how AI processes visual information, from breaking images into tiny pixels to understanding shapes and objects and then taking action.



Objectives

01

Define

Computer Vision (CV) and explain why seeing is hard for computers.

02

Describe

The CV framework: **See** → **Process** → **Act**.

03

Explain

Key concepts: pixels, features, patterns, predictions, action.

04

Identify

Common CV tasks (image classification, object detection, facial recognition, OCR).

05

Recognise

Challenges in CV (lighting, pose, occlusion, bias).

06

List

Real-world CV applications (self-driving cars, medical imaging, social media, agriculture, security).

07

Analyse

How CV works in self-driving cars and social media filters.

08

Add

A CV-focused prompt to your Prompt Library.

Recap: What We Learned in Session 14

Key Idea	What It Means
NLP	AI that reads and understands human language.
Break → Understand → Generate	The three-stage framework for processing text.
Tokenization, Lemmatization, NER	Ways to break down and label words.
Key insight	NLP is pattern matching, not true understanding.

 **Today we move from reading text to "seeing" images – Computer Vision.**

Warm-Up: The Visual Puzzle

Look at these three situations. What makes each one tricky for a computer?

Situation	Why it's hard for AI
A photo of your friend wearing a hat vs the same friend without a hat.	The face shape changes - AI might think it's a different person.
A stop sign on a sunny day vs the same stop sign covered in snow.	Colours and edges are hidden - AI may not recognise it.
A cat sitting on a patterned carpet.	AI might learn the carpet pattern instead of the cat.

Discussion: Humans see objects, faces, and signs easily even when conditions change. But computers see only numbers. How could a machine ever figure out what's in a picture?

📝 **Big Question:** How does AI turn a grid of numbers (pixels) into a meaningful label like "stop sign" or "my friend's face"?



What is Computer Vision?

Computer Vision (CV) is the branch of AI that gives computers the ability to see, interpret, and understand visual information – images and videos.

Think of CV as a **translator between the camera and the computer**. You look at a photo and instantly see a “cat” or a “stop sign”. The camera captures only a grid of coloured dots (pixels). CV breaks those pixels into patterns, compares them to what it has learned, and outputs a label or an action.

Where you see CV every day:



Face Unlock

Face unlock on your phone



Social Filters

Snapchat / Instagram filters that track your face



Google Lens

Reading text from images



Self-Driving Cars

Recognising pedestrians and signs

Key insight: CV does not "see" like you do. It finds mathematical patterns in pixels, then guesses what those patterns represent.



Pixels: The Tiny Building Blocks

Pixel (short for "picture element") is the smallest, fundamental unit of a digital image or display, appearing as a tiny, individual dot or square of color on a screen. Millions of these pixels are arranged in a grid (matrix) to form the images on televisions, smartphones, and computer monitors.

You see...	AI "sees"...
A red bicycle	A grid of numbers – each pixel has red, green, blue values (e.g., R=200, G=50, B=50 for a reddish pixel).
A smiling face	Thousands of pixels with different brightness and colour values.
A moving soccer ball	The same group of pixels changing position over time.

Pixels in Action

Think of a digital image like a mosaic artwork made of thousands of tiny coloured tiles. Each tile is a **pixel**, the tiles form a picture. AI looks at each tile individually, uses filters to detect patterns like edges, textures, or shapes and then figures out the whole image.

Just like a mosaic, every digital image is made of tiny coloured squares. The more pixels, the sharper the image – and the more data the AI has to work with.

Red Pixel

(255, 0, 0) Maximum red, no green, no blue.

Dark Blue Pixel

(0, 0, 100) No red, no green, some blue.

White Pixel

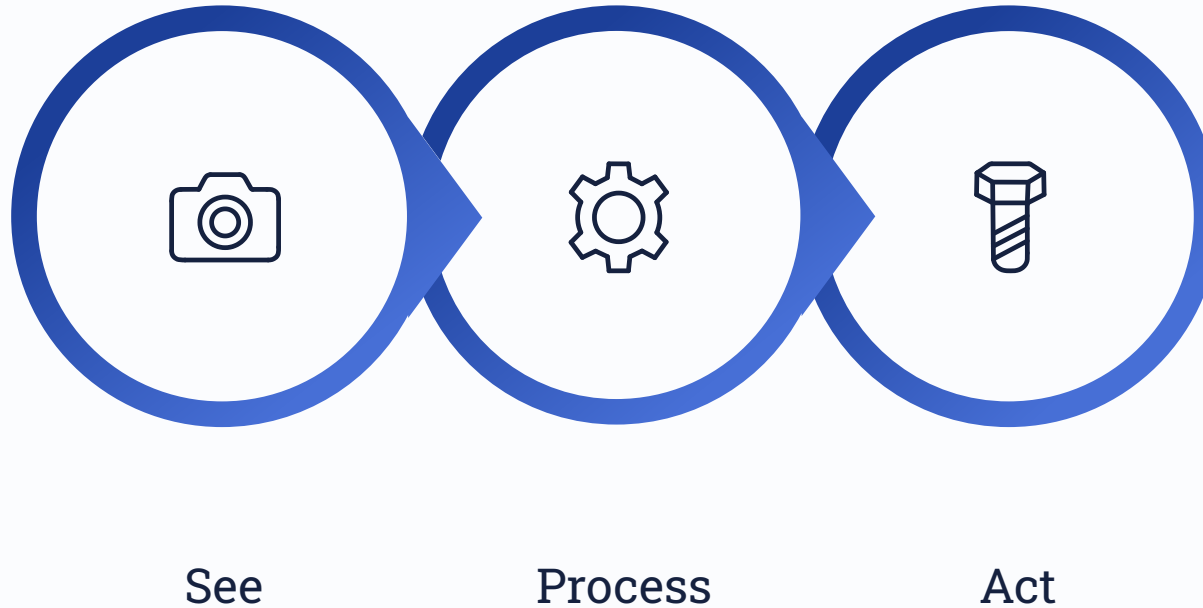
(255, 255, 255)
Maximum of all three channels.

Black Pixel

(0, 0, 0) No colour at all – pure darkness.

The CV Framework: See → Process → Act

CV follows a simple three-stage framework that matches what you learned in Session 1 (Sense → Think → Act).



The CV Framework: See → Process → Act

Let's define the stages and the key ingredients inside **Process**.

Stage	What happens	Key concepts inside
See	Capture image as pixels.	Pixels - raw numbers.
Process	Extract clues, find patterns, make a decision/prediction.	Features → Patterns → Prediction
Act	Output result or trigger an action.	Action

Inside the "Process" Stage

Term	What it means in CV	Example (face recognition)
Features	Important clues extracted from pixels (edges, corners, colours, shapes).	"Two dark spots in the upper half" → eyes. "A curved line below nose" → mouth.
Patterns	Combinations of features that repeat across many examples.	Eyes + nose + mouth arranged in a certain way → "face pattern".
Prediction	The AI's guess based on learned patterns (also called a decision).	"90% confident – this is a face."

See → Process → Act in Action

Finding Your Friend in a Crowded Cafeteria: You **See** a crowd of people (**pixels**). You look for clues: red backpack, curly hair, glasses (**features**). When you spot all three together (**pattern**), you **predict** it's your friend. Then you **act** - you wave and call their name.

📄 **How they connect:** See (Pixels) → Process (Features → Patterns → Prediction) → Act (Action)



See → Process → Act in Action

A self-driving car recognising a stop sign.

Stage	What happens	Detailed breakdown
See	Camera captures image as pixels.	Millions of pixels (raw numbers).
Process	Extract features, recognise patterns, make a prediction.	Features: red colour, straight edges, octagon shape, letters S-T-O-P. Pattern: matches training examples of stop signs. Prediction: "95% confident – this is a stop sign."
Act	Trigger an action.	Apply brakes.

❏ **Important:** The "Process" stage is pattern matching, not true understanding. The AI doesn't know what "stop" means - it has learned that certain pixel patterns usually appear together with the word "stop sign".

Mini Activity 1 : Match the CV Concept

Instructions: For each description, write which CV concept it matches: **Pixel, Feature, Pattern, Prediction, Action**.

#	Description	Concept
1	The AI's guess based on learned patterns, often with a confidence score.	_____
2	An important clue extracted from pixels, such as an edge, corner, or colour patch.	_____
3	What the system does with the prediction (e.g., unlock a phone, apply brakes).	_____
4	The smallest unit of a digital image – a dot of colour.	_____
5	A combination of features that repeats across many training examples.	_____

Mini Activity 1 : Answer Key

1. Prediction

2. Feature

3. Action

4. Pixel

5. Pattern

❏ These five concepts work together in many CV system. See (Pixels) → Process (Features → Patterns → Prediction) → Act (Action).

Common Computer Vision Tasks

Each CV task follows the same **See** → **Process** → **Act** framework with different goals.



Image Classification

Labels the main object in an image.

See: pixels of a cat.

Process: detect whiskers, pointy ears → pattern of "cat" → predict "cat".

Act: show label.



Object Detection

Finds **where** objects are (bounding boxes).

See: pixels of a street.

Process: detect car-shaped patterns → predict "car at (x,y)".

Act: draw a box.



Facial Recognition

Identifies **who** a face belongs to.

See: pixels of a face.

Process: detect unique features → pattern matches "Alice" → predict "Alice".

Act: unlock phone.



Optical Character Recognition (OCR)

Reads text from images.

See: pixels of letters.

Process: detect edges forming "A", "B" → pattern matches "HELLO" → predict text.

Act: copy to clipboard.

Challenges in Computer Vision

Even with advanced AI, CV faces many challenges. Understanding these helps you know when to trust CV and when to be careful.

Challenge	What it means	Real-world example
Lighting	Same object looks different in bright sun vs shadow.	A stop sign at night might not be recognised.
Pose & Angle	Object looks different from the side, top, or bottom.	A face from the side vs front - different features and patterns.
Occlusion	Part of the object is hidden.	A person wearing sunglasses - eyes are hidden, features missing.
Background clutter	Too many other objects confuse the AI.	A dog on a patterned carpet - AI might learn the carpet pattern instead of the dog.
Bias	Training data doesn't represent everyone equally.	Facial recognition works worse on darker skin tones if training data was mostly light skin.
Scale	Object size varies (close vs far).	A car far away is tiny - fewer pixels, features become blurry.

📌 **Key takeaway:** CV is powerful but **not perfect**. Lighting, angle, bias, etc can cause mistakes. Always test CV systems in the conditions where they will be used.

Mini Activity 2 : Spot the CV Challenge

Instructions: For each real-world scenario, identify which CV challenge is present. Choose from: **Lighting, Pose & Angle, Occlusion, Background clutter, Bias, Scale.**

#	Scenario	Challenge
1	A self-driving car fails to recognise a stop sign because the sign is partly covered by a tree branch.	_____
2	A facial recognition system works well for people with light skin but often fails for people with darker skin.	_____
3	A smartphone camera recognises a friend's face from the front, but not when the friend turns their head sideways.	_____
4	A road sign detection AI works perfectly during the day but misses signs at night.	_____
5	An AI trained to recognise cats only on a red sofa gets confused when the same cat sits on a green sofa.	_____
6	A pedestrian detection system fails to see a person who is very far away - the person is only a few pixels tall.	_____

Mini Activity 2 : Answer Key

#	Answer	Explanation
1	Occlusion	Part is hidden by the tree branch.
2	Bias	Training data doesn't represent everyone.
3	Pose & Angle	Face looks different from the side.
4	Lighting	Darkness changes features.
5	Background clutter	AI learned the wrong pattern - the sofa.
6	Scale	Object too small / far away.

❏ **Discussion:** Real CV systems must handle all these challenges. Engineers test their models in many conditions (night, side views, partial hiding) to make them robust.

Computer Vision in the Real World

Here are major areas where CV is used today. Each follows the CV framework.



Self-Driving Cars

Detects lanes, pedestrians, traffic signs. *See: pixels of road → detect white line features → predict lane position → steer.*



Medical Imaging

Finds tumours, fractures in X-rays. *See: pixels of lung scan → detect irregular blob features → predict "fracture" → alert doctor.*



Social Media Filters

Tracks facial landmarks to overlay effects. *See: pixels of face → detect eye and nose features → predict where to put bunny ears → overlay filter.*

Computer Vision in the Real World



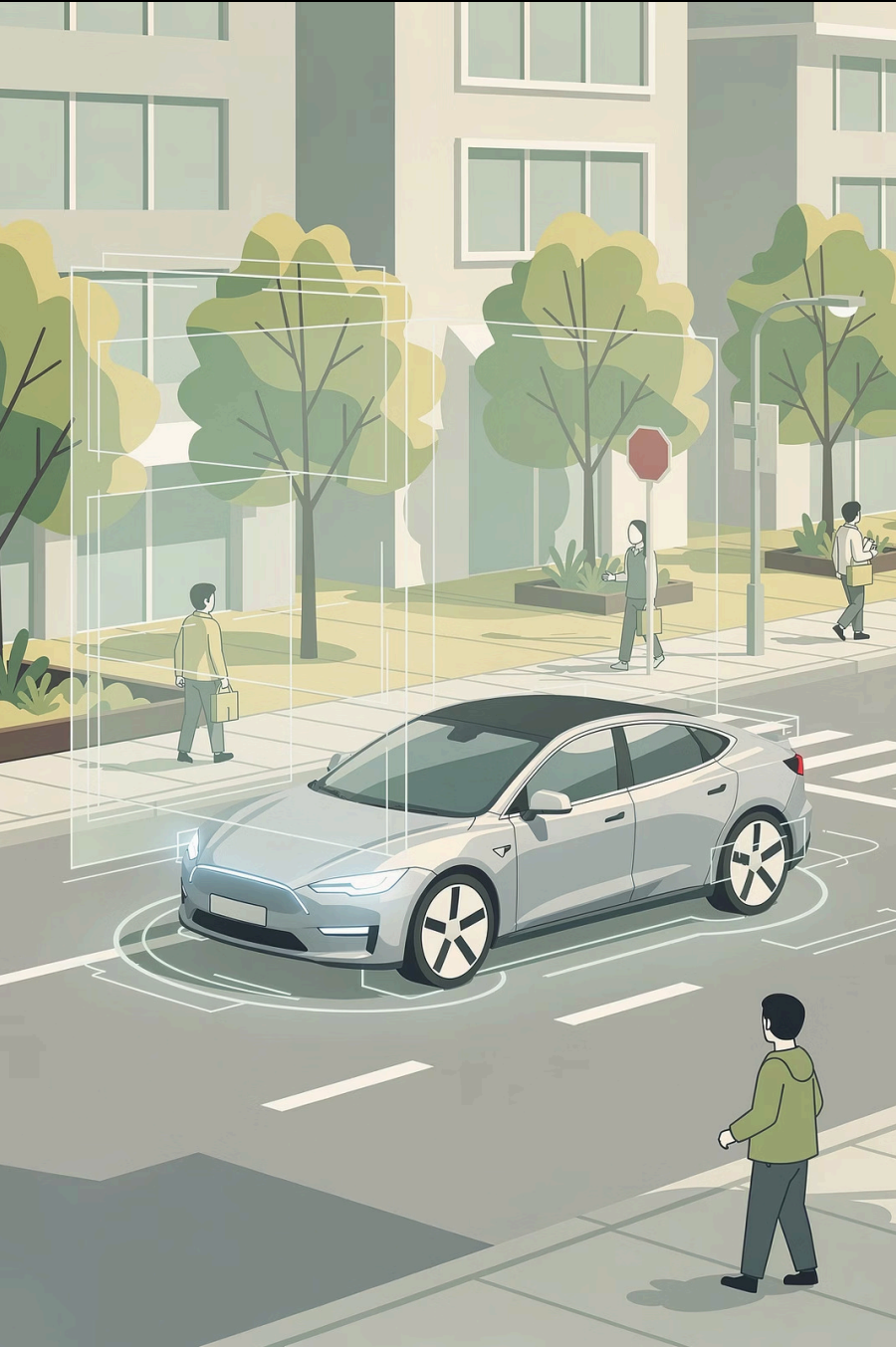
Agriculture

Monitors crops for diseases. *See: pixels of drone image* → detect yellow/brown spots → predict infected area → spray only that zone.



Security

Recognises faces, reads licence plates. *See: pixels of plate* → detect letter/number features → predict plate number → open gate.



Hands-On Activity: Real-World Computer Vision Analysis

Goal : Act as a computer vision engineer. For two real-world CV applications - **Self-Driving Cars** and **Social Media Filters** - you will identify the **features**, **patterns**, **predictions**, and **actions** for objects the AI must recognize.

Part A: Self-Driving Car Vision

A self-driving car's camera must recognise three things: **Stop Sign**, **Pedestrian**, and **Traffic Light (Red)**.

For each one, fill in the table:

CV Stage	Stop Sign	Pedestrian	Traffic Light (Red)
See (what does it look like? Describe)			
Features (list 2–3 clues)	1. 2. 3.	1. 2. 3.	1. 2. 3.
Pattern (how do features combine?)			
Prediction (what is it?)			
Act (what should the car do?)			

Part A: Sample Answers

CV Stage	Stop Sign	Pedestrian	Traffic Light (Red)
See	Red, octagon shape, white letters	Person shape, two legs, two arms, head	Red circle/arrow, on a pole, glowing
Features	Red colour, 8 straight edges, letters S-T-O-P	Vertical body, horizontal limbs, head at top	Red colour, circular shape, brightness
Pattern	Red + octagon + S-T-O-P = stop sign	Vertical + horizontal limbs + head = person	Red + circle + brightness = red light
Prediction	Stop sign	Pedestrian	Red traffic light
Act	Apply brakes	Stop and wait	Stop completely

Part B: Social Media Filter

A social media filter must track your face to add bunny ears. For it to work, the AI must recognise: **Face**, **Eyes**, and **Mouth**.

For each one, fill in the table:

CV Stage	Face	Eyes	Mouth
See (what does it look like? Describe)			
Features (list 2–3 clues)	1. 2. 3.	1. 2. 3.	1. 2. 3.
Pattern (how do features combine?)			
Prediction (what is it?)			
Act (what should the filter do?)			

Part B: Sample Answers

CV Stage	Face	Eyes	Mouth
See	Oval shape, two dark spots, curved line below	Two dark ovals, white around them, above nose	Horizontal curved line, below nose, between cheeks
Features	Oval outline, two eyes, one nose, one mouth	Dark circle, white surround, eyelid curve	Curved line, pink/red colour, horizontal orientation
Pattern	Oval + two eyes + nose + mouth = face	Dark circle + white surround = eye	Horizontal curved line + pink colour = mouth
Prediction	Face	Eye	Mouth
Act	Place bunny ears on top of face	Track eye position to keep ears aligned	Detect smile to change filter effect

Reflection Questions

Question 1

Which object was easiest to analyse?
Why? (What features were most obvious?)

Question 2

For the self-driving car, what challenge (lighting, occlusion, bias, etc.) could make it fail to recognize a pedestrian? Give an example.

Question 3

For the social media filter, what challenge could make it fail to track your eyes? Give an example.

Question 4 - Ethics Connection

If a self-driving car's pedestrian detection system was trained mostly on adults, what might happen when it sees a child? Why is that a problem?

Add to Your Prompt Library

Your New Prompt

"Explain how a self-driving car recognizes a red traffic light. Use the See → Process → Act framework and list three features."

Why This Prompt Works

- It asks for a **specific real-world application**
- It requires use of the **See → Process → Act framework**
- It asks for **three concrete features** - making the answer structured and detailed
- You can adapt it for any CV application (pedestrian, stop sign, face, etc.)

Summary

Key Term	What It Means
Computer Vision (CV)	AI that interprets images and videos.
Pixel	Smallest unit of an image - a dot of colour.
Feature	Important clue extracted from pixels (edge, corner, colour patch).
Pattern	Combination of features that repeats across examples.
Prediction	AI's guess based on learned patterns (with confidence score).
Action	What the system does with the prediction.

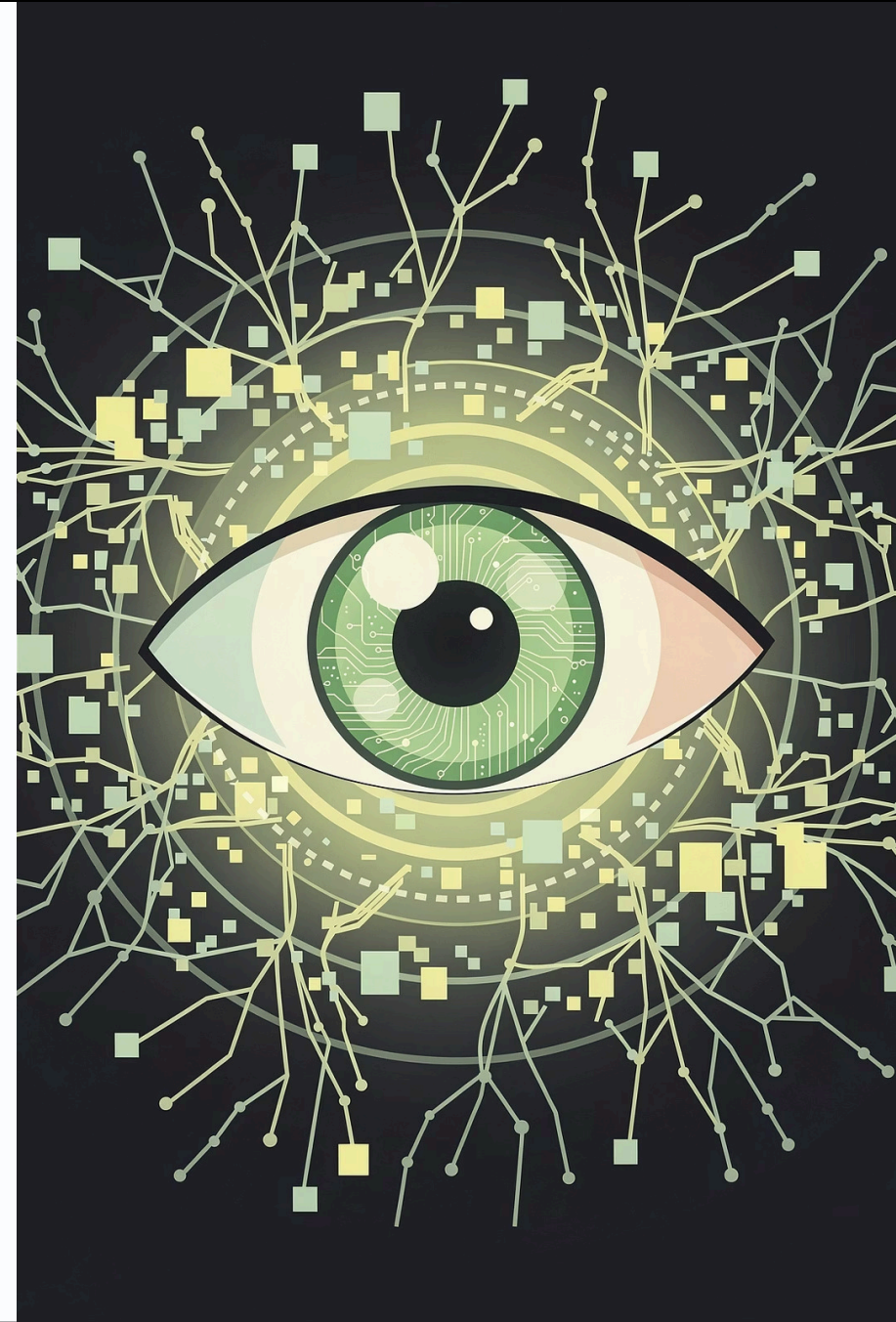
Summary

Key Term	What It Means
CV framework	See (pixels) → Process (Features → Patterns → Prediction) → Act.
Common tasks	Image classification, object detection, facial recognition, OCR.
Challenges	Lighting, pose, occlusion, bias, scale.
Real-world use	Self-driving cars, medical imaging, social media, agriculture, security.

The Most Important Idea

Computer Vision (CV) is a framework that sees pixels, processes them into features and patterns, makes a prediction, then acts

- ❑ But it's all **pattern matching, not true understanding**. That's why you must always test CV systems in different conditions and be aware of bias.



Homework

Part 1: Quick Matching

Match each term to its correct description.

#	Term
1	Pixel
2	Feature
3	Pattern
4	Prediction
5	Action

Letter	Description
A	The AI's guess based on learned patterns
B	What the system does with a prediction
C	A combination of features that repeats across examples
D	The smallest unit of a digital image
E	An important clue extracted from pixels (edge, colour, shape)

Homework

Part 2: Identify the CV Stage

For each scenario, write **See**, **Process**, or **Act**.

Scenario	Stage
A camera captures a photo of a red apple.	_____
The AI detects a round shape and red colour, then predicts "apple".	_____
The AI displays the label "apple" on the screen.	_____

Part 3: Challenge Spotting

Read the sentence and identify which CV challenge it shows: **Lighting, Pose, Occlusion, Bias, Scale**.

"A self-driving car fails to recognise a stop sign because it is covered in snow." → _____

Part 4: Add to Your Prompt Library

Add a prompt like:

"Explain how a self-driving car uses computer vision to recognise a stop sign. Use the See → Process → Act framework."

Next Session Preview

Session 16: AI That Hears & Speaks - Voice AI

Now that you've seen how AI reads (NLP) and sees (CV), next we'll explore how AI listens and speaks. You'll learn about speech recognition, text-to-speech, speech-to-text, and how tools like ElevenLabs generate synthetic voices

